IBRAHIM KHAN

ML Engineer | Conversational AI | Agentic Systems | RAG

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Summary

ML Engineer with hands-on experience architecting production-grade conversational AI systems, multi-agent pipelines, and RAG applications. I am capable for delivering measurable results: 44% API latency reduction, F1 score improvement from 0.72 → 0.81, and MRR@5 of 0.76 on live retrieval systems. Comfortable across the full stack — from LLM orchestration and vector databases to FastAPI deployment and React dashboards. Seeking to bring real-world, impact-first AI development to a client-delivery environment.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, RAG Systems, Prompt Engineering, Agentic Pipelines, OpenAI/Anthropic/Groq/Gemini APIs, Fine-tuning, Chain-of-Thought Prompting
Vector & Memory Systems: ChromaDB, Pinecone, HuggingFace Embeddings, Semantic Search, BM25, Three-Tier Memory (Ring Buffer / SQLite / Vector), NetworkX Knowledge Graphs
Backend & APIs: FastAPI, Flask, WebSocket, SSE Streaming, Redis Streams, PostgreSQL, Docker, Nginx, async SQLAlchemy
ML & Data Science: Supervised/Unsupervised Learning, Deep Learning, Feature Engineering, Hyperparameter Tuning, Pandas, NumPy, Scikit-learn, SQL, MongoDB
Frontend & Tooling: React (Vite, Zustand, TailwindCSS), Next.js, TypeScript, Python (Advanced), JavaScript, Git/GitHub, Power BI, Streamlit

EXPERIENCE

AI/ML Software Engineer Intern

Kingscote Informatics Pvt. Ltd. | *August 2025 – January 2026*

- Developed end-to-end data preprocessing pipeline for e-commerce product catalog handling 50K+ records, implementing automated feature engineering, outlier detection, and normalization that improved model F1 score from 0.72 to 0.81
- Built and deployed Flask REST API for ML model inference serving 200+ daily requests, implementing response caching and batch processing that reduced average API latency from 3.2s to 1.8s (44% improvement)
- Performed exploratory data analysis on customer behavior dataset (150K records, 25 features), uncovering 3 key user segments through clustering analysis that informed product team's feature roadmap for Q4 2024
- Collaborated with 2 senior engineers to establish model evaluation framework using cross-validation and holdout testing, creating reusable evaluation scripts now used across 4 internal ML projects

PROJECTS

Alfred — Local Multi-Agent AI Assistant (Active Development)

Python | LangGraph | FastAPI | ChromaDB | SQLite | Redis Streams | Ollama | React | Faster Whisper

- Designed a production-grade multi-agent system: LangGraph Supervisor orchestrating 5 specialist agents (Coder, Planner, Research, Memory, Echo) with a three-tier memory architecture — in-process ring buffer, async SQLite key-value store, and ChromaDB vector store — enabling persistent, context-aware conversations
- Built FastAPI gateway (sync, SSE streaming, WebSocket) with Redis Streams as event bus decoupling voice, UI, and observability layers; implemented full voice I/O pipeline — wake word detection, Faster Whisper STT, Edge TTS, tkinter HUD overlay
- Applied production engineering patterns: Abstract Base Classes for agents/tools, Factory pattern for LLM provider swapping, async-first design, and a pytest suite covering agents, routes, and all three memory tiers

Brother-Eye - Autonomous Job Agent

Python | LangGraph | FastAPI | Docker | Vector Databases (ChromaDB/Pinecone)

- Architected a 10-stage agentic pipeline — resume parsing, multi-site scraping (10 platforms), RAG-based semantic matching via ChromaDB + HuggingFace embeddings, and LLM-powered resume tailoring with anti-fabrication validation using Groq's Llama 3.3-70B
- Implemented Selenium auto-apply across 5 platforms with anti-detection measures and rate limiting; built React dashboard (Vite + TailwindCSS + Zustand) with live WebSocket scraping progress and Recharts tracking; deployed as single Docker container with Nginx

ProjectDNA — Enterprise Knowledge Intelligence Platform

Python | LangGraph | FastAPI | ChromaDB | PostgreSQL | NetworkX | Gemini 2.5 | Next.js | TypeScript

- Built a full-stack Corrective-RAG platform: self-correcting LangGraph agent grades retrieved context, rewrites poor queries and retries, then enriches answers from an auto-extracted NetworkX entity-relationship knowledge graph — all streamed live via SSE
- Designed token-aware chunking pipeline (tiktoken, 512 tokens) with Gemini embeddings (1536-dim); FastAPI backend with 8 endpoints; Next.js frontend with 4 interaction modes (Ask, Agent, Graph, Timeline); Docker Compose infrastructure (PostgreSQL 16, ChromaDB, Redis 7)

EDUCATION

Bachelor of Technology in Computer Science Engineering
11/2021 - 07/2025

Shadan College of Engineering & Technology

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Statistics & Probability, Linear Algebra
Academic Projects: Jute pest Identification and Detection (CNN, OpenCV) - achieved 97% recall on imbalanced dataset - Neural Network implementation from scratch in NumPy for MNIST classification

CERTIFICATIONS

- **Prompt Engineering for ChatGPT –** DeepLearning.AI (Andrew Ng), 2025
- **Machine Learning Specialization –** Vodafone Idea Foundation, 2024
- **Data Science Bootcamp –** Code With Harry, 2025